

Increases in Risk

Let F and G be cdf's on $\mathbb{R}_+$ with bounded support.

Definition. G is **riskier** than F if

$$\int f(z)dG(z) \leq \int f(z)dF(z)$$

for all real-valued, continuous, concave functions f on $\mathbb{R}_+$, with a strict inequality for at least one such function. (G is **strictly riskier** than F if the inequality is strict for all strictly concave functions f .)

A few implications of this definition (in each case, G is riskier than F).

- $\int zdG(z) = \int zdF(z)$. [Why? Take $f(z) = z$ to conclude that $\int zdG \leq \int zdF$ and $f(z) = -z$ to conclude that $\int zdG \geq \int zdF$.]
- $\int u(z)dG(z) \leq \int u(z)dF(z)$ for all increasing, concave vN-M utilities u on $\mathbb{R}_+$.
- $\sigma_G^2 \geq \sigma_F^2$. [Why? Let Recall that $\sigma_F^2 = \int (z - \mu_F)^2 dF = \int z^2 dF - \mu_F^2$. Apply the definition with $f(z) = -z^2$.]
- The definition extends immediately to the case of random vectors z in $\mathbb{R}_+^n$. Recall that you proved that the value function $V(p) = \max_{a \in A} \sum_{i=1}^n u(a, i)$ is *convex* in p . Suppose that p^0 is a decision maker's *prior* belief about which of the n states of the world will be realized, that is, what beliefs would be if without any further information. Now suppose that the decision maker can observe a signal that takes on one of two values: "high" with probability $\lambda \in (0, 1)$ and "low" with complementary probability. And suppose that the *posterior* belief after seeing the high signal would be p^h and after observing the low signal it would be p^ℓ . On average these beliefs equal the prior belief: $p^0 = \lambda p^h + (1 - \lambda)p^\ell$. The convexity of V implies that expected utility with some information is higher than expected utility with no information: $V(p^0) \leq \lambda V(p^h) + (1 - \lambda)V(p^\ell)$. In words, *the expected value of information is nonnegative*. More generally *the riskier the distribution of posterior beliefs, the higher is the decision maker's expected utility*. This suggests that a criterion for one signal being more informative than another is that the distribution of posterior beliefs is riskier for the more information signal.¹

Exercise. Consider the two-risky asset problem from the Handout, Pratt's Theorem on Risk Aversion and Portfolio Demand, with a C^3 vN-M utility u with $u' > 0$, $u'' < 0$. (a) Show that DARA implies $u''' > 0$. (b) Suppose that $u''' > 0$. Show that a sufficient condition for an decrease in risk to increase the demand for the risky asset is that $-u'''(z)z/u''(z) < 2$ (the ratio $-u'''(z)z/u''(z)$ is known as **relative prudence**).

Exercise. Extend the conclusion of MWG 6.C.9 to the case in which the distribution G of y replaces F and G is strictly riskier than F (that is, the starting distribution is not necessarily degenerate.)

¹Blackwell, David. "Equivalent comparisons of experiments." *The Annals of Mathematical Statistics* 24, (1953): 265-272.